

1.) Throughout a time interval, while the speed of a particle increases as it moves along the x axis, its velocity and acceleration might be:

- A. negative and negative, respectively
- B. positive and negative, respectively
- C. positive and zero, respectively
- D. negative and zero, respectively
- E. negative and positive, respectively

Answer: A. negative and negative, respectively

2.) An engineer named Cindy Vortex is designing the runway of an airport. If the plane that will use the airport has an acceleration rate is likely to be 3 m/s^2 , the takeoff speed for this plane will be 65 m/s . Assuming this minimum acceleration, what is the minimum allowed length for the runway?

- A. 226 m
- B. 704 m
- C. 1216 m
- D. 550 m

Answer: B. 704 m

3.) This term refers to the speed of an object throughout its whole journey.

- A. Instantaneous speed
- B. Average speed
- C. Average velocity
- D. Instantaneous velocity

Answer: B. Average speed

4.) Danny Phantom starts dashing passing through walls at 15 m/s and after a while, its speed changes to 45 m/s . How many seconds have passed if his acceleration is 3 m/s^2 ?

- A. 8 s
- B. 10 s
- C. 9 s
- D. 7 s

Answer: B. 10 s

5.) If Heihachi Mishima vertically leaps reaching a height of 1.29 m , then what is his takeoff speed?

- A. 22.62 m/s
- B. 73 m/s
- C. 12.85 m/s
- D. 5.03 m/s

Answer: D. 5.03 m/s

6.) A stone is thrown vertically upward with an initial speed of 19.5 m/s . It will rise to a maximum height of:

- A. 9.8 m
- B. 38 m
- C. none of these
- D. 19.4 m
- E. 4.9 m

Answer: D. 19.4 m

7.) A feather, initially at rest, is released in a vacuum 12 m above the surface of the Earth. Which of the following statements is correct?

- A. The acceleration of the feather decreases until terminal velocity is reached
- B. The maximum velocity of the feather is 9.8 m/s during the fall
- C. The acceleration of the feather increases
- D. The acceleration of the feather is zero
- E. The acceleration of the feather remains constant during the fall

Answer: E. The acceleration of the feather remains constant during the fall

8.) As a rocket is accelerating vertically upward at 9.8 m/s^2 near Earth's surface, it releases a projectile. Immediately after release, the acceleration (in m/s^2) of the projectile is _____.

- A. none of the above
- B. 19.6 up
- C. 9.8 up
- D. 0
- E. 9.8 down

Answer: E. 9.8 down

9.) The position vs. time graph above shows the motion of Reyna as she went from Site A to Site B. However, she felt tired at some time intervals she went stationary. At which time interval was she at rest?

- A. 0 s
- B. 15 s - 30 s
- C. she was never at rest during her journey
- D. 5 s - 15 s

Answer: D. 5 s - 15 s

10.) The position vs. time graph above shows the motion of Omen as he went from Site A to Site B. What was his instantaneous velocity at $t = 20\text{s}$?

- A. 1.08 m/s
- B. 0.47 m/s
- C. 1.08 m/s
- D. -0.47 m/s

Answer: D. -0.47 m/s

11.) The position vs. time graph above shows the motion of Omen as he went from Site A to Site B. What was his instantaneous velocity at $t = 10\text{s}$?

- A. 0 m/s
- B. 2.42 m/s
- C. 0.47 m/s
- D. 1.08 m/s

Answer: A. 0 m/s

12.) Which of the following is true about the position vs. time graph of an object shown above?

- A. The object is slowing down.
- B. The object is moving at a speed of 0 m/s.
- C. The object has positive acceleration.
- D. The object has negative acceleration.

Answer: B. The object is moving at a speed of 0 m/s.

13.) The velocity of a projectile equals its initial velocity added to:

- A. constant vertical velocity
- B. a constantly increasing downward velocity
- C. a constant horizontal velocity
- D. constantly increasing horizontal velocity
- E. a constant velocity directed at the target

Answer: B. a constantly increasing downward velocity

14.) Velocity is defined as:

- A. change of position
- B. rate of change of acceleration with time
- C. speeding up or slowing down
- D. rate of change of position with time
- E. position divided by time

Answer: D. rate of change of position with time

15.) A plane traveling north at 200 m/s turns and then travels south at 200 m/s. The change in its velocity is:

- A. zero
- B. 400 m/s north
- C. 200 m/s north
- D. 400 m/s south
- E. 200 m/s south

Answer: D. 400 m/s south

16.) Identical guns fire identical bullets horizontally at the same speed from the same height above two planets, one on the Earth and one on the Moon. Which of the following three statements are true?

- I. The horizontal distance travelled by the bullet is greater for the Moon.
- II. The flight time is less for the bullet on the Earth.
- III. The velocities of the bullets at impact are the same.

- A. I and II only
- B. I, II, III
- C. III only
- D. I and III only

Answer: A. I and II only

17.) For a biological sample in 1.0-m radius centrifuge to have a centripetal acceleration of 25g its speed must be:

- A. 245 m/s
- B. 122 m/s
- C. 16 m/s
- D. 11 m/s
- E. 50 m/s

Answer: C. 16 m/s

18.) A Toyota and Honda are making a turn. The Toyota is four times more massive than the Honda. If they make the turn at the same speed, which of the following is true about their centripetal forces?

- A. The centripetal force on the Toyota is half that of Honda.
- B. The centripetal force on the Toyota is four times greater than that of Honda.
- C. The centripetal force on the Toyota is two times greater than that of Honda.
- D. The centripetal force on the Toyota is eight times greater than that of Honda.

Answer: B. The centripetal force on the Toyota is four times greater than that of Honda.

19.) Which of the following is true about the relationship of centripetal force and an object moving in uniform circular motion?

- A. The centripetal force of an object moving in UCM is always perpendicular with the motion of the object.
- B. The centripetal force of an object moving in UCM is always coinciding with the motion of the object.
- C. The centripetal force of an object moving in UCM is always parallel with the motion of the object.
- D. The centripetal force of an object moving in UCM is always anti-parallel with the motion of the object.

Answer: A. The centripetal force of an object moving in UCM is always perpendicular with the motion of the object.

20.) What happens to the centripetal acceleration of an object in uniform circular motion if its velocity doubles?

- A. The centripetal acceleration will triple.
- B. The centripetal acceleration will quintuple.
- C. The centripetal acceleration will quadruple.
- D. The centripetal acceleration will double.

Answer: C. The centripetal acceleration will quadruple.

21.) An object has an initial velocity of 3 m/s after 10 seconds it stopped at 25 meters distance from its initial position. What is the final velocity of the object?

- A. 1 m/s
- B. 2 m/s
- C. 4 m/s
- D. 3 m/s

Answer: B. 2 m/s

22.) A pen and a rock are dropped vertically at the same time with the same height. Neglecting air resistance which object will hit the ground the ground first?

- A. Not enough information
- B. Both the pen and the rock
- C. The rock
- D. The pen

Answer: B. Both the pen and the rock

23.) Natasha Romanoff tossed a coin vertically upward and reaches a maximum height of 0.5 m before it comes back down. With what velocity was it thrown?

- A. 4.63 m/s
- B. 4.82 m/s
- C. 4.08 m/s
- D. 4.16 m/s

Answer: C. 4.08 m/s

24.) Clint Barton shot an arrow straight up with an initial velocity of 30.5 m/s. Find the time to reach the highest point.

- A. 5.1 seconds
- B. 5.4 seconds
- C. 3.1 seconds
- D. 3.22 seconds

Answer: C. 3.1 seconds

25.) What is the total distance travelled of Olivia if she is walking at a speed of 5km/hr in 25 secs?

- A. 5 m
- B. 137.16 m
- C. 173.61 m
- D. 125 m

Answer: D. 125 m

26.) An object has an initial velocity of 3 m/s after 10 seconds it stopped at 25 meters distance from its initial position. What is the acceleration of the object?

- A. 0.3 m/s^2
- B. -0.3 m/s^2
- C. -0.1 m/s^2
- D. 0.1 m/s^2

Answer: C. -0.1 m/s^2

27.) A ball player catches a ball 14.63 seconds after throwing it vertically upward. With what speed did he throw it?

Answer: 71.69

28.) A race car accelerates uniformly from 16.27 m/s to 21.58 m/s in 11.77 seconds. Determine the acceleration of the car.

Answer: 0.45